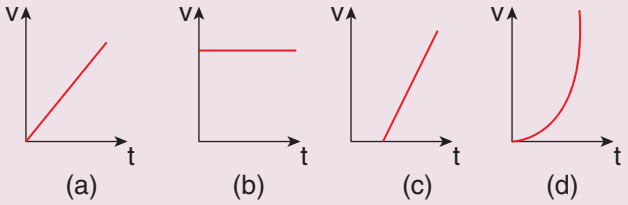
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maximum height. What is the angle of projection of the object?

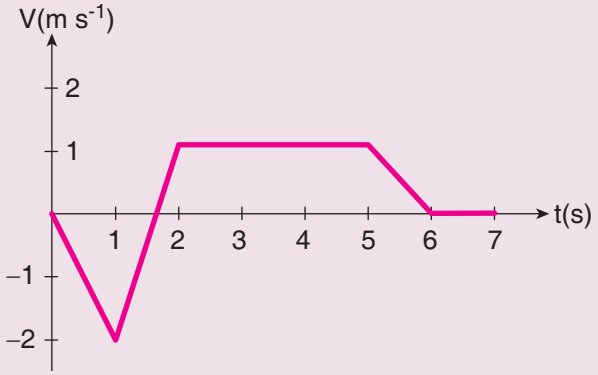
[Ans: $\theta = 45°$]

7. The following graphs represent velocity – time graph. Identify what kind of motion a particle undergoes in each graph.



[Ans: (a) $\vec{a} = constant$ (b) $\vec{v} = constant$ (c) $\vec{a} = constant$ but greater than first graph (d) $\vec{a}$ *is variable*]

8. The following velocity–time graph represents a particle moving in the positive x–direction. Analyse its motion from 0 to 7 s. Calculate the displacement covered and distance travelled by the particle from 0 to 2 s.



[Ans: distance=1.75 m, displacement = -1.25 m]

9. A particle is projected at an angle of θ with respect to the horizontal direction. Match the following for the above motion.

(a) v_x – decreases and increases

(b) v_y – remains constant

(c) Acceleration – varies

(d) Position vector – remains downward

[Ans: v_x = *remains constant*, v_y = *decreases and increases*, a = *remains downward*, r = *varies*]

10. A water fountain on the ground sprinkles water all around it. If the speed of the water coming out of the fountain is v. Calculate the total area around the fountain that gets wet.

[Ans: Area = $\frac{\pi v^4}{g^2}$]

11. The following table gives the range of a particle when thrown on different planets. All the particles are thrown at the same angle with the horizontal and with the same initial speed. Arrange the planets in ascending order according to their acceleration due to gravity, (g value).

Planet	Range
Jupiter	50 m
Earth	75 m
Mars	90 m
Mercury	95 m

[Ans: $g_{jupiter}$ is greater, $g_{mercury}$ is smaller]

12. The resultant of two vectors A and B is perpendicular to vector A and its magnitude is equal to half of the magnitude of vector B. Then the angle between A and B is

a) 30° b) 45°

c) 150° d) 120°

[Ans: $\theta = 150°$]

13. Compare the components for the following vector equations